Operation and Maintenance Manual of Temperature (Humidity) Test Chamber

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I. Also called environmental test equipment, temperature (humidity) test chamber is able to test the heat, cold, dry and humidity resistant performance of all kinds of materials. It is applied to the quality test of such materials as electronics, electric apparatus, telecommunications, instrument, vehicle, plastics, metal, food, chemicals, building materials, medical treatment and aerospace.

II. Performance Specification:

1. **Model: TH - 800-A**
2. **Temperature scope: 25℃ ~150℃**
3. **Humidity scope: 20%R.H. ~ 98%R.H.**

   Note: Please refer to the drawing about the application scope of temperature and humidity!!
4. **Temperature and humidity control precision: Temperature: ±0.5℃; Humidity: ±2.5%RH.**

   Note: The performance guarantee of the 2nd~4th items shall be tested 30 minutes after the temperature sensing becomes stable!
5. **Distribution uniformity of temperature and humidity:**

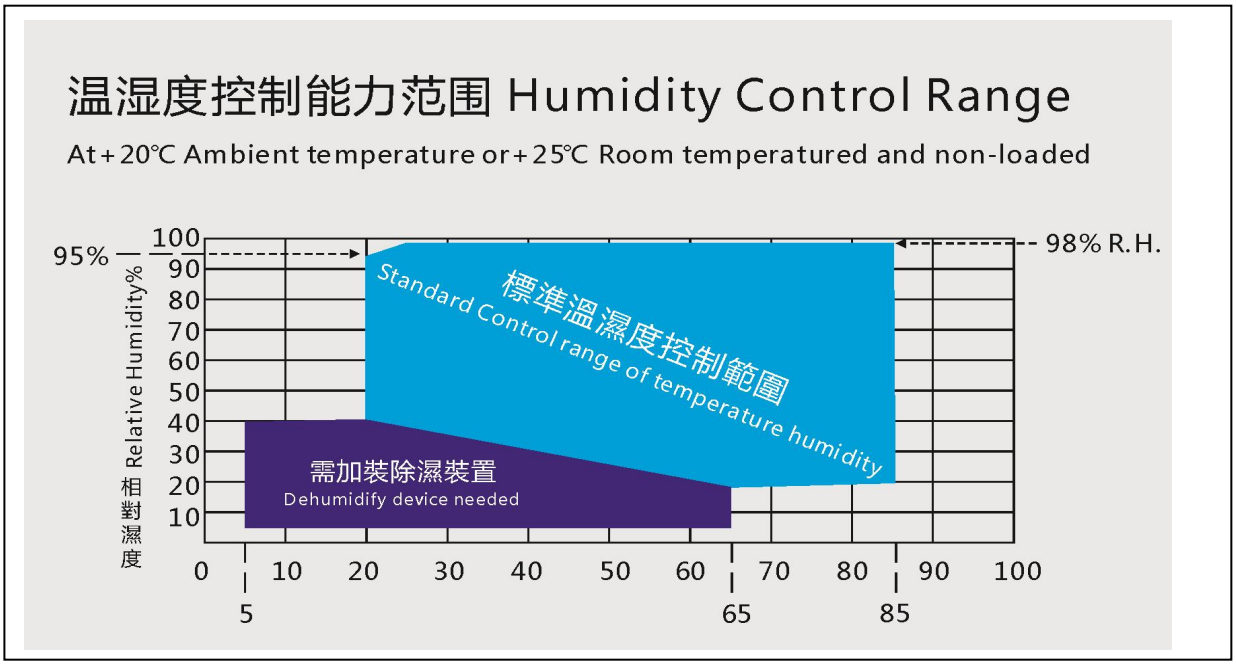
   Temperature: ±2.0℃; Humidity: ±3.0%

   Note: The performance guarantee of the aforesaid 5 items needs measuring 15cm from each wall surface of the test area. The performance shall be measured 30 minutes after the temperature and humidity becomes stable!
6. **Temperature rise velocity: 3.0℃/minute; when it is under empty load and non linear control)**
7. **Cooling velocity: 1.0℃/minute; when it is under empty load and non linear control)**
8. **Size of inner box: W1000 x H1000x D800 mm**
9. **Space for equipment installation: At least 700MM shall be left at the right, left and back side of the equipment and 1,600MM shall be left for the front side.**

10. **Door to test box: 1 (except for non-standard machine type).**

11. **Power: AC380V±10% 50/60HZ 3 phase 5-wires**

13. **Drawing about the control range of temperature and humidity:**



III. System Structure:

1. **Refrigeration system:**

   a. Imported highly effective full-closed or semi-closed low-noise compressor is provided.

   b. Fin radiating motor condenser.

   c. Fin multi-stage automatic load volume adjustment evaporator.

   d. Environmentally friendly refrigerant R404A or R23

   e. Other accessories: Drying agent, oil separator, high-voltage protection switch, refrigerant flow window, etc.

   f. Expansion system: Original European and American imported components are used integrally.

2. **Electronic system (safety protection system):**

   a. ZCP theorist power controller.

   b. Compressor overheat protection switch.

   c. Compressor overflow protection switch.

   d. Humidifier quick fuse.

   e. Heater quick fuse.

   f. Separation protection switch.

   g. Water shortage protection switch.

3. **Air duct system:**

   a. Specially customized long stainless steel axis motor is provided.

b. Multi-blade stainless steel cage accelerates heat and humidity circulation volume.

4. **Heating system: Fin stainless steel electrical heated tube is provided.**
5. **Humidifying system: Electric heating evaporation U-shaped stainless steel humidifier is provided.**
6. **Sensing system:Two Pt100 are used and both temperature and humidity are measured by A/D switch through the comparison and input of dry and wet bulb.**
7. **Water system:**

   a. Built-in stainless steel water tank.

   b. Automatic water supply device (water is pumped from lower to upper layer).

   c. Water shortage indication & alarm. Alarm in case of short-term water shortage and shutdown in case of long-term water shortage.
8. **Control system:**

   Temperature and humidity controller: Original imported TATO-5166 LCD touch screen programmable temperature and humidity controller and LCD controller.

a. Controller specification:

- **Precision: Temperature ±0.1℃＋1digit, humidity ±1%R.H＋1digit**
- **Resolution: Temperature ±0.1. Humidity ±0.1%R.H.**
- **Temperature slope: 0.1~9.9can be set.**
- **Upper and lower limit standby and alarming function**
- **Temperature and humidity input signal is PT 100Ω×2 (dry and wet bulb)**
- **Temperature and humidity conversion output: 4~20MA**
- **6 groups of P.I.D. control parameter settings and P.I.D. automatic calculation**
- **Automatic correction of dry and wet bulb**

b. Picture display function:

- **Contact data input and direct screen display**
- **Temperature and humidity setting (SV) and direct display of actual value (PV)**
- **The number of program executed, segment, rest time and circulation times can be displayed.**
- **Accumulative operation time**
- **The setting values of temperature and humidity are displayed by graph curve and program curve execution function (for color) is displayed in time.**
- **Independent program editing picture is provided and temperature, humidity and time can be input directly.**

- **Both Chinese and English display.**
- **C. Program capacity and control function: 99 PATTEN in maximum**
- **Applicable memory volume: 100 SEGMENTS**
- **Orders can be executed repeatedly: Each order can be executed as many as 50 times.**
- **Program is made interactively and is capable of editing, removing and inserting.**
- **SEGMENTS time settings 530Hour**
- **It is capable of power-off memory so that it can restart automatically and continue executing the program after power-on.**
- **USB communication interface (software sold additionally)**
- **The graph curve (for color) can be displayed in time when executing the program.**
- **It is capable of start up and shutdown by presetting.**
- **It is capable of date and time adjustment.**
- **It is capable of key-pressing and picture lock**

9. Chamber material:

a. Material of inner box: SUS#304 heat and cold resistant stainless steel plate

b. Material of outer box: SUS#304 Stainless steel powder coated

c. Insulation material: Imported high-strength PU foaming insulation material

IV. Working Principle:

1. Temperature adjustment and humidity control system controls SSR through PID so that the heating and humidification amount of the system is equal to heating and humidity consumption.

2. It will be input to controller CPU by A/D conversion through temperature measurement signal of dry and wet bulb and then output to I/O plate by comparing with RAN. I/O plate issues order to make air supply system and refrigeration system work. PID will also control or heat or humidify SSR to make the heating and humidification amount come to the test box through air supply system evenly in order to adjust temperature and humidity evenly.

VI: Preparations:

The purchaser shall be liable for getting prepared for all the contents below before using this tester!

Power: □AC380V 3 Phases 5lines 50/60HZ Please refer to the requirements for water, electricity and gas of the equipment

Note: Ensure the range of fluctuation of voltage frequency of this tester: Voltage

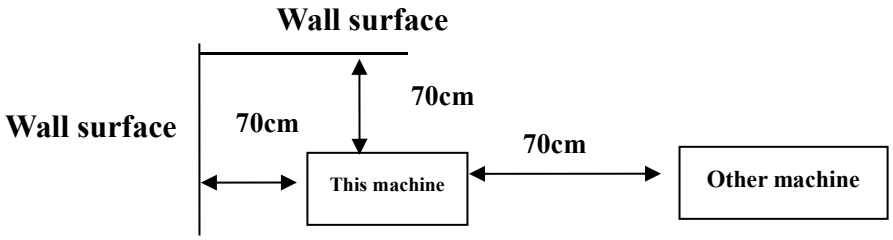
±5%; frequency ±1%!

Humidification water consumption: Pure water, distilled water (the primary consumption shall be above 30L) or the water, the electric conductivity of which is below 10μs/cm can be used only.

Note: The purer the better! Underground water is prohibited!

VII. Installation Site and Method:

1. **The place where this tester is installed shall be able to radiate heat of this tester and easy to test and maintain.**
2. **Although this tester is made of senior stainless steel plate, parts and electronic controller parts are easily affected unfavorably if the tester is installed in humid place, resulting in dis-operation or accelerated aging**
3. **The tester shall be installed on the solid and horizontal ground where drainage device is installed. Good ventilation of the surrounding environment is required and at the bottom of machine is refrigeration system, the heat of which is very considerable. Please install it at least 70cm from the external environment, walls, between any other machines and the machine so as to ensure ventilation, as shown in the figure below:**



4. **The tester shall be set on the flat ground where no vibration happens. Four horizontal pins at the bottom of machine shall be installed and positioned so as to balance the tester, facilitate drainage inside the tank and prevent abnormal noise.**
5. **The tester shall be installed where direct sunshine can be avoided, in order to avoid affecting test accuracy; indoor air circulation and air temperature shall be maintained (air conditioner in laboratory is strongly recommended).**
6. **The tester shall be installed where no other electrical equipment or inflammable and explosives are set in order to avoid hazards. Please put the machine in isolated room instead of near public place, in-flammables, explosives or perishable chemicals to avoid fire and personal injury if failure happens.**
7. **Please never install this tester where it is dirty and dusty. The tester can not be operated most stably until the surrounding temperature and humidity are 10℃~25℃ and 70±10%RH respectively; otherwise, The cooling speed of this tester is slow or unable to reach the low temperature requirements and the temperature control is unstable and has large fluctuation.**

8. It is prohibited to put any article on the top of the tester to avoid injuring persons and damaging articles after it falls.

9. Never use portable electric box, wire or motor as the supporting point of driving force when handling it in order to avoid accidental failure due to the damage and loosening of electric box and apparatus.

10. The maximum inclination of box shall be below 30° and the fix shall be firmed to avoid falling, damaging by pressing or injuring person or damaging articles.

11. If the well positioned tester needs moving, take away four horizontal pins before movement; then well adjust them to avoid damaging the tester.

VIII. Power Configuration and Installation Method:

Please distribute power according to the method below. Pay attention to power capacity. Never share one external power for several machines to avoid a lower voltage and thus affecting machine performance and even causing failure shutdown. Please use special loop instead.

1. **The power supply circuit shall be as short as possible. If the power voltage is too low, please install pressure stabilizing system to reach voltage rated value; otherwise, the system can not be started due to too low voltage and as a result, motor and controller will be burnt.**
2. **Please arrange power line according to the specification table:**

	1∅2W220VAC		**50HZ**
	3∅3W200VAC		**60HZ**
	3∅3W380VAC		
	3∅5W380VAC		

3. **The diameter of applied power is: (The cable is within 10m)**

	2.0~2.5mm^2		**8.0~10.0mm^2**
	3.5~4.0 mm^2		**14~16 mm^2**
	5.5~6.0mm^2		**22~25mm^2**

4. **For 3-phase power, please pay attention to the sequence and out-of-phase protection (the phase may be reserve if 3-phase power has electricity but the machine does not work. If so, change two adjacent power lines). Make sure earthing is set (to avoid hazards if electricity leakage happens).**
5. **If you connect earthing line with water tube, water tube must be earthing metal pipe (not all the metal pipes can be grounded effectively).**
6. **Never damage wiring during installation! Never connect earthing line with gas pipe.**
7. **Before power configuration, please check if machine and power line are**

damaged and machine is deformed during the transportation, if air supply circulation is complete and if the inner box is clean and tidy.

8. **Please confirm the configuration of power line carefully.**
9. **The range of fluctuation of power voltage as input by machine can not be beyond the allowable scope and the earthing line must be good. Otherwise, it may affect the performance of this machine.**
10. **Please configure proper safety device according to machine power so that the power can be cut off safety in case of machine failure in order to avoid fire disaster and personal injury.**
11. **Please arrange wires after the machine is well positioned in space place and ensure the consistency of wire and rated current and voltage of machine; otherwise, electric shock and accident may happen.**
12. **The wiring personnel shall be professional enough to avoid connecting wires by mistake or damaging machines and parts by inputting error power.**
13. **Please arrange wire after ensuring the input power is cut off to avoid electric shock.**
14. **If the machine is provided with 3-phase motor, please check if the rotation direction is correct when connecting power. The rotation direction of single phase motor has been well adjusted before delivery. It is necessary to check if the rotating direction is correct when changing it so as to avoid affecting machine performance.**
15. **After wiring is finished, input the power which coincides with it on the basis that the control electrical apparatus of machine has no failure. Install all the electric box covers before power-on; otherwise, electric shock and fire disaster may happen.**
16. **Anyone but full-time professionals is not allowed to maintain and check this tester; this tester must be dismantled and checked after power-off to avoid electric shock and fire disaster.**
17. **It is not allowed to work by dismantling electric box, door to electric box, side plate of machine and some safety protection devices since the machine is under dangerous operation status.**
18. **Do not operate the main power switch on control panel too frequently. Shut down the power switch of machine or the user's air switch only when shutting down this tester.**

IX. Attentions Before Usage:

a. Never start this machine if it is moved intensively. Put it statically for 2 hours before start up after it is moved.

b. Confirmation of power line and earthing line:

Is the power line properly connected and grounded according to the specification?

c. Are the body and enclosure damaged?

d. Is operation motor blocked?

e. Is air door blocked?

When this tester is started for the first time or after the humidification water box is cleaned, please set the temperature 3℃ lower than room temperature and set humidity as 1%RH. Operate for 30minutes after start up before setting other temperatures and humidity.

f. Confirmation of over temperature protector of dry and wet bulb:

● Temperature setting point of over temperature protector

g. Confirmation of water supply:

● Is the water contained in water tank reservoir enough?

● If the cover of water tank covered properly?

● Has the drainage switch been reset?

h. Confirmation of drainage pipe:

●Is drainage pipe connected properly?

i. Confirmation of wet bulb test cloth:

● Is it clean? [clean it once a month and change a new one for every three months]

●Is the position correct? [put it in the front end of humidity sensor!]

● Is the water level of feed water tank normal?

● Is it humid? [please change new test cloth if water remains in tank and water level is normal but test cloth is not wet]

j. Confirmation of humidifier water level:

● Is water level normal? [check if the water level of water level control box is normal 3-5minues after the humidifier supplies water.】

● Is humidifier water clean? [the water addition tray of humidifier shall be drained regularly and cleaned by brush to make it clean] 】

k. Check if the over temperature gas protection switch of wet bulb is set correctly.

l. Confirmation of drainage pipe:

●Is drainage pipe connected properly?

Note: Take out of test cloth if temperature is tested only. If test cloth can not be operated under 85℃, change another new one before operation next time; otherwise, water may not be absorbed. The new test cloth shall be washed by clean water; otherwise, it easily fails to absorb water.

Fix horizontal pins properly after all the items above are confirmed.

X. Operating Procedure:

1. **Start user's air switch, main power switch of equipment [NFB] and then power key in sequence.**
2. **Set the program to be executed according to the specification as long as normal picture is displayed on temperature and humidity instrument, in order to set the test conditions smoothly.**
3. **The setting value of temperature, humidity and time shall coincide with the requirements of user's tested sample.**
4. **The functions of temperature and humidity instrument shall be set by full-time professionals or according to the operating manual of temperature and humidity instrument.**
5. **Close the tester door after finishing setting all functions. This tester will be operated normally after the door is closed.**
6. **Pay attention to the rotation direction of operator motor and 3-phase motor when the machine operates initially.**
7. **The machine will operate according to the temperature and humidity test conditions by pressing the operation switch of controller.**
8. **If the machine needs shutting down after the temperature and humidity reach the user's product requirements, the product can not be taken out of until the switch of temperature and humidity gauge of controller is closed.**
9. **Please refer to controller operating manual for the settings of relevant program and the usage method of fixed set point control; see the specification of temperature and humidity gauge for the specific operation of temperature and humidity gauge.**
10. **This machine is provided with view observant lamp. Please light the light within the box by pressing the switch of observant lamp if checking the situations inside the box.**
11. **In case of failure, the display screen of this machine will display the position where failure happens immediately and the buzzer will alarm. Please press [RESET] key on the display after failure is removed and then restart this tester.**

XI. Operation Attentions:

1. **This machine is prohibited to heat or test explosive, combustible or corrosive substances. Otherwise, it may cause unnecessary loss or the machine is no longer able to test it.**
2. **The amount of test articles can not affect air volume circulation of test box;**

otherwise, it may affect machine performance.

3. **Close the door closely when the machine is occupied; otherwise, temperature and humidity may leak and are unable to reach the performance region.**
4. **This machine shall be operated by the designated personnel to avoid damaging this machine earlier.**
5. **The machine is provided with test hole at the left side and it can be used for testing the circuit inside the box (TS signal power interface is set, test power inside the box can be connected and the maximum current is 5A.)**
6. **Open the light switch within box if trying to see the changes within the box during the test. See the change by the window.**
7. **Try to avoid opening the door to the box if the machine operates below 0℃ since this will cause ice blockage of internal evaporator and other parts. The lower the temperature, the more serious the ice blockage will be. If the door must be opened, the shorter, the better.**
8. **When this machine is operated with low temperature, it is a MUST to set 40℃ drying treatment for half an hour and open the door to the box to avoid next operation. The evaporator will be frozen or the test article will be damaged. (If failure happens to the machine by failing to set 40℃drying treatment but the machine is within the warranty period, we will bear no liability for free maintenance since such failure is artificial).**
9. **Never open the door to the box during operation unless it is a necessity; otherwise, the following unfavorable consequences may happen:**

   High temperature and humidity gas rush out of the box...very dangerous!

   The intern side of door to box is still in high temperature... causing injury!

   High-temperature air may trigger fire alarm and result in dis-operation!

   Method to open the door: The operators must move backward along the direction of door-opening when opening the door to prevent massive of hot gas inside the machine from injuring persons.
10. **Please ensure this machine is grounded safely and correctly to avoid electrostatic induction!**
11. **Please check circuit breaker, temperature and over temperature protector, test articles and operator's safety protection on a regular basis.**
12. **The correct relative humidity can be obtained by installing test cloth correctly.**
13. **Never inspect it during the operation by hands to avoid electric shock or injury by fan. Please shut down the machine and cut off the power prior to repairing.**
14. **Anyone other than full-time professionals is not allowed to maintain or inspect this machine. Maintenance shall be carried out at the witness of supervisor to avoid**

electric shock by switching on without informing relevant personnel.

15. **Please read controller specification and relevant manuals carefully before operating this machine.**

XII. Maintenance Attentions and Method:

1. **The radiator (condenser) of refrigerating unit shall be maintained and cleaned regularly. The condenser shall be maintained on a monthly basis since high-voltage switch may trip and thus produce false alarm if the condenser is covered by dusts. The vacuum cleaner, hard brush or high-pressure air jet shall be used to remove the dusts attached on the radiating meshes of condenser.**
2. **It is not allowed to make test object contact the upper trimmings of door when opening/closing the door or taking the object from the box so as to avoid damaging the trimmings and shortening the service life.**
3. **The surroundings and bottom ground of the machine shall be clean and tidy to prevent massive dusts absorbed in the unit from accidents and reducing performance.**
4. **Refrigerating system plays a pivotal role in this machine so it is of necessity to check copper pipes semiannually to see if it leaks refrigerant and check the joints of horns and welded junction. For any oil leakage, please notify us immediately or deal with it directly.**
5. **The large current junction of switchboard. The distribution room shall be cleaned and overhauled once a year at least. The loosening of junction may make the machine under dangerous working state.**
6. **The humidifier and water tray shall be cleaned frequently to avoid dirt and affecting steam emission. They are generally cleaned after test is finished. The timely removal of dirt can extend the service life of humidification pipe and ensure the smoothness of water flow. Copper brush shall be used for cleaning and sufficient water shall be used for flushing.**
7. **The wet cloth of dry and wet bulb shall be checked frequently. Change the test cloth if it does not absorb water when the surface of test cloth is not clean or becomes hard or test cloth is controlled above 85℃ or before temperature and humidity control is made, since it is related to the accuracy of humidification test. Test cloth shall be changed in two or three months in general. When changing test cloth, clean cloth shall be used to wipe the temperature sensing element. Washing by water is required when changing new test cloth and water tank shall be cleaned thoroughly.**
8. **Check and adjustment of water level of wet bulb**

The water level of water collection basin can not be too high since water overflows

easily. Too low water level of water collection basin makes water absorption of wet bulb test cloth abnormal and it thus affects the accuracy of wet bulb. The water level is proper if it is 60% or 70% of the total depth. The water level of water collection basin can be adjusted by adjusting the height of water collection box [adjust the screws on both sides]

9. The setting value of two over temperature protectors within the distribution box can not be debugged arbitrarily since they have been debugged well prior to delivery. This protection switch is used for the alarming of air separation and water shortage of heating pipe and humidification pipe. The setting point= temperature setting point +20℃~30℃.

10. When the temperature within the test box rises to the setting point of over temperature protector when this machine is operated, the power supply of heater will be stopped and [OVER HEAT] over temperature alarming light will be on but the fan still operates. If the machine is operated in the long run but no one supervises it, please check if the over temperature protector is well set before operation.

11. Check and maintenance of water circuit

The water pipe of water circuit is easily to be blocked or leak. Please check it regularly. If any blockage or leakage, please remove it immediately or notify us. The water inside the humidifier shall be changed once a month to ensure clean water and smooth water flow.

12. When taking out of product after test is due, the working personnel must, after this machine is shut down, wear dry, anti-power and temperature resistant gloves before operation;

13. Cleanliness and maintenance inside and outside the body

Prior to operation, the sundries within the machine shall be removed thoroughly.

The distribution chamber shall be cleaned by cleaner once a year at least so as to remove the dusts inside it. The external part of box must be cleaned by wiping it with WD-40 anti-rust oil on a quarterly basis at least.

14. We bear no liabilities if loss occurs by arbitrarily refitting this machine without our permission during the application.

15. We bear no liabilities for all the consequences caused by the failure in abiding by the operation and attentions.

XIII. Attached Accessories:

1. **Rail track, the height of which is adjustment is set within the test box.**
2. **Stainless steel supporter.**
3. **Wet bulb gauze for standard test.**

4. **Standby fuse.**
5. **Operation and maintenance specification and operating manual of temperature and humidity control.**

XIV. Troubleshooting:

Fault	Cause	Troubleshooting
The machine does not take action after power-on	Phase shortage or abnormal power	Check it and transmit power again
	The controller has no output	Check if controller works
	The fuse of controlling power is burnt	Check the cause and change fuse
	Malfunction of safety protection device	Check safety protection measures and reset manually
Failure to cool down or slow cooling	Refrigerating compressor is broken	Change or overhaul it
	Refrigerant leakage	Overhaul it by notifying us or professionals
	Operation motor does not rotate or rotates reversely	Check the cause or upgrade it
	The electromagnetic switch of compressor is broken	Check the cause or upgrade it
	Condenser becomes dusty	Blow the dusts away by high-pressure nozzle
	Poor radiation of unit	Improper installation or re position and re-install it.
Temperature fails to reach the required value or is out of control	Temperature over temperature protection action	Reset it
	Temperature SSR is damaged.	Check and change it
	Temperature electrical heated tube is broken	Check and change it
	Temperature contactor is broken	Check and change it
	Failure of refrigeration system	Overhaul it by notifying us or professionals
Humidity fails to reach the required value or is out of control	Humidity SSR is damaged.	Check it and change a new one
	Humidity heating tube is broken	Check it and change a new one
	Water absorption effect of wet cloth is poor	Change a new one
	Too high or too low water level of water tray	Adjust water level
	Humidity contactor is broken	Check the cause or upgrade it
	Temperature over humidity protection action	Reset it
Controller is out of control	Key pressing does not work	Notify us
	No power input	Check it and transmit power again

	I/O plate does not work	**Line bank becomes loose and reinsert it**
	Control panel has no display	**Check power or notify us**